Murakkab shakllarning yuzi va perimetri

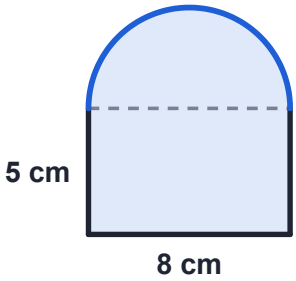
1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Rasmdagi shakl to'g'ri to'rtburchak va yarim doiradan iborat. Shakl perimetrini topishda qaysi chiziq qo'shilmaydi?**



A) Pastki tomon (8 cm)
B) Punktir chiziq
C) Yarim aylana yoyi
D) Yon tomonlar (5 cm)

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2. **Kvadratdan chorak doira olib tashlandi. Qolgan bo'yalgan soha yuzi qanday topiladi?**

A) chorak doira yuzi − kvadrat yuzi
B) kvadrat yuzi − chorak doira yuzi
C) kvadrat yuzi + chorak doira yuzi
D) kvadrat perimetri − chorak yoy uzunligi

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3. **Radiusi bir xil bo'lgan 4 ta chorak doira birga qo'yilsa, nima hosil bo'ladi?**

A) bitta butun doira **B)** yarim doira
C) ikkita doira **D)** kvadrat

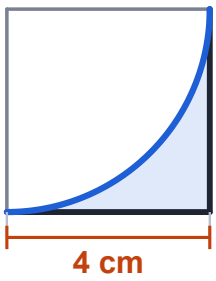
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4. **Radiusi 5 cm bo'lgan ikkita bir xil yarim doiraning umumiy yuzini π orqali ifodalang.**

A) 50π cm² **B)** 25π cm² **C)** 12,5π cm² **D)** 10π cm²

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5. **Kvadratning tomoni 4 cm. Rasmdagi bo'yalgan soha yuzini toping (π = 3,14).**



A) 3,44 cm² **B)** 4,56 cm²
C) 12,56 cm² **D)** 28,56 cm²

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6. **Shakl to'g'ri to'rtburchak (6 m × 3 m) va uning uzun tomoniga chizilgan yarim doiradan iborat (diametri 6 m). Shakl yuzini toping (π = 3,14).**

A) 46,26 m² **B)** 27,42 m² **C)** 32,13 m² **D)** 74,52 m²

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7. **Eshik to'g'ri to'rtburchak (eni 10 dm, bo'yi 4 dm) va uning ustidagi yarim doiradan iborat (diametri 10 dm). Eshik perimetrini toping (π = 3,14).**

A) 29,7 dm **B)** 49,4 dm **C)** 33,7 dm **D)** 43,7 dm

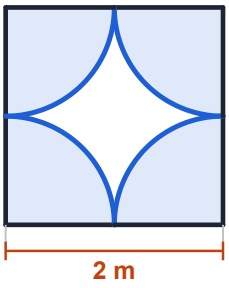
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8. **Chorak doiraning radiusi 4 cm. Undan ikki tomoni radius bo'lgan to'g'ri burchakli uchburchak olib tashlandi. Qolgan qism yuzini toping (π = 3,14).**

A) 4,56 cm² **B)** 8 cm² **C)** 12,56 cm² **D)** 20,56 cm²

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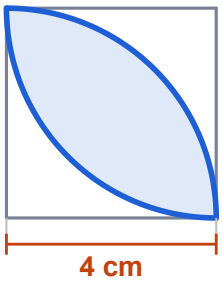
9. **Kvadratning tomoni 2 m. Uning har bir burchagida bir xil chorak doira bor, ular bir-biriga tegib turadi (rasm). Bo'yalgan chorak doiralarning umumiy yuzini toping (π = 3,14).**



A) 3,14 m²
B) 0,86 m²
C) 12,56 m²
D) 6,28 m²

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10. **Kvadrat plitkaning tomoni 4 cm. Unga yaproq naqsh chizilgan: ikkita chorak aylana, markazlari plitkaning qarama-qarshi uchlarida (rasm). Naqsh chegarasining uzunligini toping (π = 3,14).**



A) 12,56 cm
B) 25,12 cm
C) 6,28 cm
D) 28,56 cm

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11. **Stol usti to'g'ri to'rtburchak (10 dm × 8 dm) va uning ikki qisqa tomoniga chizilgan ikkita yarim doiradan iborat (diametri 8 dm). Stol usti yuzini toping (π = 3,14).**

A) 105,12 dm²
B) 280,96 dm²
C) 180,48 dm²
D) 130,24 dm²

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12. **Doira 3 ta teng bo'lakka bo'lindi. Doiraning radiusi 3 dm. Bitta bo'lakning yuzini toping (π = 3,14).**

A) 28,26 dm²
B) 14,13 dm²
C) 9,42 dm²
D) 6,28 dm²

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13. **Doira 3 ta teng bo'lakka bo'lindi. Doiraning radiusi 12 cm. Bitta bo'lakning perimetrini toping (π = 3,14).**

A) 37,12 cm
B) 25,12 cm
C) 61,12 cm
D) 49,12 cm

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14. **To'g'ri to'rtburchak shaklidagi maydon 10 m × 4 m. Uning qisqa tomonidan diametri 4 m bo'lgan yarim doira shaklidagi joy ajratildi (maydon ichkarisiga). Qolgan maydon yuzini toping (π = 3,14).**

A) 46,28 m²
B) 14,88 m²
C) 33,72 m²
D) 27,44 m²

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15. **Kvadratning tomoni 6 cm. Uning bir tomoniga tashqaridan yarim doira chizildi (diametri — kvadrat tomoni). Shakl perimetrini π orqali ifodalang.**

A) (24 + 3π) cm
B) (18 + 3π) cm
C) (18 + 6π) cm
D) (24 + 6π) cm

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16. **Kvadratning tomoni 6 cm. Uning bir tomoniga tashqaridan yarim doira chizildi (diametri — kvadrat tomoni). Shakl yuzini π orqali ifodalang.**

A) (36 + 4,5π) cm²
B) (36 + 9π) cm²
C) (24 + 4,5π) cm²
D) (36 + 18π) cm²

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17. **Kvadratning bir tomonidan yarim doira kesib olinib, qarama-qarshi tomonga tashqaridan yopishtirildi. Shaklning yuzi qanday o'zgaradi?**

A) kamayadi
B) o'zgarmaydi
C) ikki marta ortadi
D) ortadi

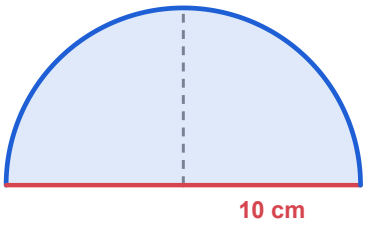
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18. **Tomoni 4 cm bo'lgan kvadrat kartonning chap tomonidan diametri 4 cm bo'lgan yarim doira qirqildi va o'ng tomoniga tashqaridan yopishtirildi. Yangi shaklning perimetrini toping (π = 3,14).**

A) 33,12 cm **B)** 16 cm **C)** 20,56 cm **D)** 28,56 cm

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19. **Shakl radiusi 10 cm bo'lgan ikkita bir xil chorak doiradan iborat (rasm). Shakl yuzini toping (π = 3,14).**



A) 157 cm² **B)** 31,4 cm²

C) 314 cm² **D)** 78,5 cm²

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20. **Kvadratning tomoni 6 m. Uning bir tomoniga tashqaridan radiusi 6 m bo'lgan chorak doira yopishtirildi (chorak doiraning bir radiusi — kvadrat tomoni). Shakl yuzini toping (π = 3,14).**

A) 64,26 m² **B)** 46,26 m² **C)** 92,52 m² **D)** 7,74 m²

Javoblar

Murakkab shakllarning yuzi va perimetri · 1-test (oson)

№	Javob	Qiymati
1	**B**	Punktir chiziq
2	**B**	kvadrat yuzi − chorak doira yuzi
3	**A**	bitta butun doira
4	**B**	25π cm²
5	**A**	3,44 cm²
6	**C**	32,13 m²
7	**C**	33,7 dm
8	**A**	4,56 cm²
9	**A**	3,14 m²
10	**A**	12,56 cm

№	Javob	Qiymati
11	**D**	130,24 dm²
12	**C**	9,42 dm²
13	**D**	49,12 cm
14	**C**	33,72 m²
15	**B**	(18 + 3π) cm
16	**A**	(36 + 4,5π) cm²
17	**B**	o'zgarmaydi
18	**C**	20,56 cm
19	**A**	157 cm²
20	**A**	64,26 m²

Qisqa yechimlar

1. Perimetr — shaklni o'rab turgan chiziq.
Punktir chiziq shakl ichida qoldi, u chegara emas.
Javob: **punktir chiziq**.

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2. Bo'yalgan soha — kvadratdan bo'lak olib tashlangandan keyin qolgan qism.
Demak, kattadan kichikni ayiramiz: **kvadrat yuzi − chorak doira yuzi**.

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3. Chorak — doiraning to'rtdan biri.
4 ta chorak: $4 \cdot \frac{1}{4} = 1$, ya'ni **bitta butun doira**.

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4. Ikkita yarim doira — bitta butun doira.
Yuzi: π · 5 · 5 = **25π cm²**.

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5. Kvadrat: 4 · 4 = 16 cm².
Chorak doira (radiusi 4 cm): $\frac{1}{4} \cdot 3{,}14 \cdot 4 \cdot 4 = 12{,}56$ cm².
Bo'yalgan soha: 16 − 12,56 = **3,44 cm²**.

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6. To'g'ri to'rtburchak: 6 · 3 = 18 m².
Yarim doiraning radiusi 6 : 2 = 3 m: $\frac{1}{2} \cdot 3{,}14 \cdot 3 \cdot 3 = 14{,}13$ m².
Shakl: 18 + 14,13 = **32,13 m²**.

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7. Yarim aylana yoyi: $\frac{1}{2} \cdot 3{,}14 \cdot 10 = 15{,}7$ dm.
To'g'ri chiziqlar: pastda 10 dm, yonlarda 4 + 4 dm. Ustki tomon ichkarida — qo'shilmaydi.
Perimetr: 10 + 4 + 4 + 15,7 = **33,7 dm**.

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8. Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 4 \cdot 4 = 12{,}56$ cm².
Uchburchak: $\frac{1}{2} \cdot 4 \cdot 4 = 8$ cm².
Qolgan qism: 12,56 − 8 = **4,56 cm²**.

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9. Chorak doiralar tomon o'rtasida tegadi: radius 2 : 2 = 1 m.
4 ta chorak doira — bitta doira: 3,14 · 1 · 1 = **3,14 m²**.

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10. Shakl chegarasida faqat ikkita chorak yoy bor.

Bitta yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 4 = 6{,}28$ cm.

Perimetr: $2 \cdot 6{,}28 =$ **12,56 cm**.

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11. To'g'ri to'rtburchak: $10 \cdot 8 = 80$ dm².

Ikkita yarim doira — bitta doira, radiusi $8 : 2 = 4$ dm: $3{,}14 \cdot 4 \cdot 4 = 50{,}24$ dm².

Jami: $80 + 50{,}24 =$ **130,24 dm²**.

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12. Butun doira: $3{,}14 \cdot 3 \cdot 3 = 28{,}26$ dm².

Bitta bo'lak — uning $\frac{1}{3}$ qismi: $28{,}26 : 3 =$ **9,42 dm²**.

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13. Yoy — aylananing $\frac{1}{3}$ qismi: $\frac{1}{3} \cdot 2 \cdot 3{,}14 \cdot 12 = 25{,}12$ cm.

Ikkita radius: $12 + 12 = 24$ cm.

Perimetr: $25{,}12 + 24 =$ **49,12 cm**.

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14. Yarim doiraning radiusi $4 : 2 = 2$ m: $\frac{1}{2} \cdot 3{,}14 \cdot 2 \cdot 2 = 6{,}28$ m².

Qolgan maydon: $10 \cdot 4 - 6{,}28 = 40 - 6{,}28 =$ **33,72 m²**.

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15. Yarim aylana yoyi: $\frac{1}{2} \cdot \pi \cdot 6 = 3\pi$ cm.

Kvadratning 3 ta tomoni chegarada: $6 + 6 + 6 = 18$ cm (to'rtinchisi ichkarida).

Perimetr: **(18 + 3π) cm**.

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16. Kvadrat: $6 \cdot 6 = 36$ cm².

Yarim doiraning radiusi 3 cm: $\frac{1}{2} \cdot \pi \cdot 3 \cdot 3 = 4{,}5\pi$ cm².

Shakl: **(36 + 4,5π) cm²**.

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17. Bir tomondan qancha olingan bo'lsa, ikkinchi tomonga shuncha qo'shildi.

Shuning uchun yuz **o'zgarmaydi**.

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18. Tepa va pastki tomonlar: $4 + 4 = 8$ cm.

Ikkita yarim aylana yoyi: $2 \cdot \frac{1}{2} \cdot 3{,}14 \cdot 4 = 12{,}56$ cm.

Perimetr: $8 + 12{,}56 =$ **20,56 cm**.

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19. Ikkita chorak doira — yarim doira.

Yuzi: $\frac{1}{2} \cdot 3{,}14 \cdot 10 \cdot 10 =$ **157 cm²**.

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20. Kvadrat: $6 \cdot 6 = 36$ m².

Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 6 \cdot 6 = 28{,}26$ m².

Bo'laklar qo'shildi: $36 + 28{,}26 =$ **64,26 m²**.
